Radioactivity does not interfere in the least with chemical reactions. Thus, the chemical properties of element 43 have been investigated in a whole series of experiments. The investigation of element 61 is more difficult, because it is one of the rare earth elements, and its chemical properties are almost identical with those of the other members of that group of elements.

The other two missing elements, those of atomic numbers 85 and 87, have also been produced artificially. Element 85 was produced by Corson, McKenzie and Segré from bismuth, by bombardment with alpha rays of high energy (32 Mev.),and was named *astatine* (symbol: *At*). It is formed by the following reaction:

$$_{83}Bi^{209} + {}_{2}He^{4} = {}_{85}At^{211} + {}_{0}n^{1} + {}_{0}n^{1}$$

Since that time, traces of $_{85}At^{218}$ have been detected, by Karlik and Bernert, among the natural radioactive substances, as a product of the decay of $_{84}RaA^{218}$ ($_{84}Po^{218}$).

Element 87 is a product of the decay of a neptunium isotope. Traces of it have also been detected by Perrey in the radioactive decay of natural actinium. It has been named *francium.*

Finally, the periodic system of elements, which used to end with uranium, has been extended artificially to atomic number 96. We have already mentioned neptunium and plutonium. These two elements are produced principally in the uranium reactor, in the way indicated in the following formulae:

$$_{92}U^{238} + {}_{0}n^{1} = {}_{92}U^{239} \longrightarrow {}_{93}Np^{239} + {}_{-1}e^{0}$$
$$_{93}Np^{239} \longrightarrow {}_{94}Pu^{239} + {}_{-1}e^{0}$$

But other isotopes of these elements have also been produced artificially.

Elements 95 (*americium—Am*) and 96 (*curium—Cm*) have been obtained as follows:

$$_{92}U^{238} + {}_{2}He^{4} = {}_{94}Pu^{241} + {}_{0}n^{1};$$
$$_{94}Pu^{241} \longrightarrow {}_{95}Am^{241} + {}_{-1}e^{0};\ {}_{94}Pu^{239} + {}_{2}He^{4} = {}_{96}Cm^{242} + {}_{0}n^{1}$$

The production of new chemical elements is therefore no longer a dream of the future, but an important part of modern nucleonics.